

```

sketch_oct8a.ino
1 #include <Adafruit_Fingerprint.h>
2 #include <SoftwareSerial.h>
3 SoftwareSerial mySerial(10, 11); // RX, TX for fingerprint
4 Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);
5 String usernames[15] = {
6   "Rakesh Jain", // ID 1 - Admin
7   "Anil", // ID 2
8   "Gunmala", // ID 3
9   "Rekha", // ID 4
10  "Monu", // ID 5
11  "Niket", // ID 6
12  "Garima", // ID 7
13  "Sonu", // ID 8
14  "Neetu", // ID 9
15  "Tinu", // ID 10
16  "Vikas", // ID 11
17  "Chintu", // ID 12
18  "Sarita", // ID 13
19  "Harita", // ID 14
20  "Deepak" // ID 15
21 };
22 void setup() {
23   Serial.begin(9600);
24   while (!Serial);
25   finger.begin(57600);
26   if (finger.verifyPassword()) {
27     Serial.println("Fingerprint sensor ready");
28   } else {
29     Serial.println("Fingerprint sensor not found");
30     while (1);
31   }
32   Serial.println("Enter ID (1-15) to enroll:");
33 }
34 void loop() {
35   while (!Serial.available());
36   int id = Serial.parseInt();
37   if (id < 1 || id > 15) {
38     Serial.println("Invalid ID. Enter 1-15.");
39     return;
40   }

```

Fig. 3: Code snippet for fingerprint enrollment

```

13   "Sonu", // ID 8
14   "Neetu", // ID 9
15   "Tinu", // ID 10
16   "Vikas", // ID 11
17   "Chintu", // ID 12
18   "Sarita", // ID 13
19   "Harita", // ID 14
20   "Deepak" // ID 15
21 };
22 void setup() {
23   Serial.begin(9600);
24   while (!Serial);
25   finger.begin(57600);
26   if (finger.verifyPassword()) {
27     Serial.println("Fingerprint sensor ready");
28   } else {
29     Serial.println("Fingerprint sensor not found");
30     while (1);
31   }
32   Serial.println("Enter ID (1-15) to delete:");
33 }
34 void loop() {
35   while (!Serial.available());
36   int id = Serial.parseInt();
37   if (id < 1 || id > 15) {
38     Serial.println("Invalid ID. Enter 1-15.");
39     return;
40   }
41   Serial.print("Deleting fingerprint for: ");
42   Serial.println(usernames[id - 1]);
43   deleteFingerprint(id);
44 }
45 void deleteFingerprint(int id) {
46   int p = finger.deleteModel(id);
47   if (p == FINGERPRINT_OK) {
48     Serial.println("Deleted successfully.");
49   } else {
50     Serial.println("Failed to delete fingerprint.");
51   }
52 }
53 }
54 }

```

Fig. 4: Enrollment deletion

components used are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the electronic voting system. It employs an Arduino Uno, an R305 fingerprint sensor, a 20×4 LCD display, four push buttons, and a buzzer to support a tamper-proof and user-friendly voting process. Each of the 14 authorised users (10 voters and 4 candidates) is pre-registered and permitted to vote only once.

(EFY note. A 20×4 LCD with a 4/8-bit interface has been used in this system. It may be replaced with a 16×2 I2C LCD to reduce wiring. In such

a case, the code must be adjusted accordingly. The system has also been tested with an I2C display and found to work reliably.)

After successful fingerprint authentication, the system displays the voter's name on the LCD and prompts the selection of one of four candidates: Anil, Gunmala, Rekha, or Monu. A buzzer confirms that the vote has been recorded. The vote count remains concealed until the authorised administrator (Rakesh Jain) authenticates via fingerprint, after which the final tally is displayed.

Pressing the Arduino reset button clears previous data and reinitialises the system. This prevents duplicate voting and supports transparency, making the system suitable for institutional voting, small-scale elections, and pilot e-governance implementations. Its simplicity and low hardware requirements contribute to practicality and ease of deployment.

The system operates through three main processes: Fingerprint enrolment, fingerprint deletion, and the main voting process. User fingerprints are added using the fingerprint enrolment sketch software. When removal is required, the fingerprint deletion sketch erases stored fingerprint templates.

The main voting program manages the voting sequence. The system includes one administrator (Rakesh Jain), four candidates (Anil, Gunmala, Rekha, and Monu), and ten registered voters (Niket, Garima, Sonu, Neetu, Tinu, Vikas, Chintu, Sarita, Harita, and Deepak). Each

```

42 const int buzzerPin = 12;
43 int votes[4] = {0, 0, 0, 0}; // Vote count
44 bool hasVoted[15] = {false}; // IDs 1 to 15
45 String usernames[15] = {
46   "Rakesh Jain", // ID 1 - Admin
47   "Anil", // ID 2
48   "Gunmala", // ID 3
49   "Rekha", // ID 4
50   "Monu", // ID 5
51   "Niket", // ID 6
52   "Garima", // ID 7
53   "Sonu", // ID 8
54   "Neetu", // ID 9
55   "Tinu", // ID 10
56   "Vikas", // ID 11
57   "Chintu", // ID 12
58   "Sarita", // ID 13
59   "Harita", // ID 14
60   "Deepak" // ID 15
61 };
62 void setup() {
63   pinMode(button1, INPUT_PULLUP);
64   pinMode(button2, INPUT_PULLUP);
65   pinMode(button3, INPUT_PULLUP);
66   pinMode(button4, INPUT_PULLUP);
67   pinMode(buzzerPin, OUTPUT);
68   lcd.begin(20, 4);
69   Serial.begin(9600);
70   finger.begin(57600);
71   if (finger.verifyPassword()) {
72     Serial.println("Sensor ready");
73   } else {
74     Serial.println("Sensor error");
75     while (1);
76   }
77   lcd.clear();
78   lcd.print("Place Finger...");
79 }
80 void loop() {
81   int id = getFingerprintID();

```

Fig. 5: Snippet for main voting code